MATH 332 Linear Algebra: Theory and Applications

Fred J. Hickernell | Quiz 3 | Thursday, October 15, 2026

Instructions:

i. *This quiz has **four questions**, worth **20 points**.*

ii. *The time allowed is **15 minutes**. Approved testing accommodations apply.*

iii. *This quiz is closed book, but you may use up to **4** double-sided letter-sized sheets of notes.*

iv. *No calculators or other electronic devices are allowed. Silence and put devices in your bag under your desk or at the front of the room.*

v. *Work individually. Do not communicate with other students or share materials. Direct questions to the instructor.*

vi. *Give a brief justification for each answer. A sentence or two, or a short calculation, is usually enough.*

vii. *Print your name clearly on each answer sheet.*

Q	Max	Score
1	5	
2	5	
3	5	
4	5	
Tot	**20**	

I understand these instructions and have not relied on any help for this exam beyond what is allowed, nor provided any help to anyone else beyond what is allowed.

_____ _____

Signature *Date*

_____

Printed Name

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1. (5 points) Let $\boldsymbol{u}$ and $\boldsymbol{v}$ be nonzero vectors in $\mathbb{R}^3$. Contrast the geometric meaning of $\boldsymbol{u}^\mathsf{T}\boldsymbol{v} = 0$ with that of $\boldsymbol{u} \times \boldsymbol{v} = \boldsymbol{0}$.

   Answer: A zero dot product means the vectors are perpendicular. A zero cross product means they are parallel (possibly in opposite directions). The dot product is a scalar; the cross product is a vector. The nonzero assumption excludes the case in which both products vanish simply because an input is zero.

2. (5 points) A student claims that the span of two nonzero vectors cannot contain the zero vector. Is the claim correct? Explain using the definition of span.

Answer: No. The span contains every linear combination of the two vectors, including $0\boldsymbol{u}+0\boldsymbol{v} = \mathbf{0}$. *Nonzero generators do not require nonzero coefficients in a linear combination.*

3. (5 points) Let V be the set of all continuous real-valued functions on $\mathbb{R}$ satisfying $f(0) = 1$. With the usual function addition and real scalar multiplication, is V a vector space?

Answer: No. Any one of the following reasons suffices:

- *The zero function has value* 0 *at the origin, so it is not in* V.
- V *is not closed under scalar multiplication:* $f(t) = 1$ *belongs to* V, *but* $0f$ *is the zero function and does not belong to* V.
- V *is not closed under addition: if* $f, g \in V$, *then* $(f+g)(0) = f(0)+g(0) = 1+1 = 2$, *so* $f+g \notin V$.

Students need not check every axiom.

4. (5 points) In the real vector space $\mathbb{P}_2$ of polynomials of degree at most two, does $1+2t+t^2$ belong to $\text{span}\{1+t,\ t+t^2\}$? Justify your answer.

Answer: Yes, since

$$1+2t+t^2 = 1(1+t)+1(t+t^2).$$

This expresses the polynomial as a linear combination of the two generators, with real coefficients 1 *and* 1.

Quiz 3 Scores (out of 20)
Number of Students: 20, Minimum: 0, Maximum: 20, Mean: 12.8, Median: 14
Standard Deviation: 6.4, Quartiles (Q1, Q3): (9.5, 19)

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2|00000
1|0001355678
0|00589
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